ShipGuard — Predictive Freight Intelligence Platform

Shipment Delay Risk Forecasting, Real-World Telemetry & GenAI Operations Engine

Project: ShipGuard (Fresa Technologies)	**Architecture:** React 18 + FastAPI + Python ML + Oracle / SQLite
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1. Executive Summary

In global freight forwarding, unexpected delays result in severe financial penalties (demurrage, detention) and degraded service reliability. ***ShipGuard*** *is an enterprise-grade predictive intelligence system engineered to shift logistics operations from reactive fire-fighting to proactive risk prevention. The platform calculates transit delay probabilities before vessel departure, continuously ingests real-world telemetry (weather events, terminal dwell times, public holidays, currency shifts), and utilizes Large Language Models (Google Gemini & NVIDIA NIM) to generate automated root-cause explanations and operational action plans in real-time.*

2. Business Problems & Applied Solutions

Logistics Operational Challenge	Business Impact	ShipGuard Solution
Blind Spot Exceptions Delays only discovered after containers are stranded at transshipment hubs.	High demurrage/detention fees, SLA breach penalties.	**Predictive Risk Scoring:** Forecasts delay risk (Low/Med/High) 48-72h prior to occurrence.
Disconnected Telemetry Severe weather, port strikes, and local holidays ignored by static ERPs.	Unanticipated schedule disruptions and vessel rollover.	**Live External Sync:** Dynamic risk delta adjustments based on live API signals.
Communication Overheads Operations managers spend hours writing status emails & reports.	Slow turnaround time, lost operational focus.	**GenAI Incident Co-Pilot:** One-click delay notifications, mitigation plans & reports.
Data Leakage in Traditional ML Future milestone data contaminating training features.	Overly optimistic models that fail in live production.	**Point-in-Time Queries:** Strict historical boundaries based only on data before ETD.

3. Technical Architecture & Component Stack

Layer	Technology	Architecture & Implementation Details
Frontend	React 18, Vite, Vanilla CSS	High-performance single-page app (SPA); operational dashboard with risk lane filters, modal telemetry, live metrics, and real-time AI assistant interface.
Backend API	FastAPI, Python 3.11+, Pydantic	Asynchronous RESTful architecture with automated OpenAPI documentation, strict request/response data contracts, and dependency injection.
AI / ML Layer	Scikit-Learn, NumPy, Custom Sigmoid Engine	Probabilistic feature extraction combining route baselines, carrier historical reliability, transit variances, and weather/port delta modifiers.
GenAI Services	Google Gemini API & NVIDIA NIM	Multi-turn operations co-pilot, automated natural language incident reports, structured prompt engineering with dynamic context injection.
Database	Oracle DB (Prod) / SQLite (Dev)	Dual-database architecture: Enterprise Oracle Autonomous DB schema (DDL, identity sequences, indexes) + zero-configuration SQLite for local dev/demos.
Security	JWT Auth, bcrypt, Rate Limiter	Stateless Bearer token auth, salted password encryption, and multi-tier sliding-window rate limiters preventing API abuse and quota exhaustion.

4. Core Functional Modules

A. Predictive Risk Scoring Engine

Scores each shipment continuously on a 0.0% to 99.9% probability scale, classifying shipments into actionable risk tiers: **LOW (<33%)** for standard milestone tracking; **MEDIUM (33%-65%)** for pre-emptive status alerts; and **HIGH (≥66%)** for immediate operational intervention and carrier escalation.

B. External Real-World Telemetry Ingestion

A background telemetry worker periodically pulls external environmental factors to modify baseline risk scores:

- **Port Weather:** Ingests wind speeds, storm warnings, and heavy precipitation (+4% to +12% risk delta).
- **Terminal Congestion:** Analyzes vessel queue lengths and average dwell times at transshipment ports (+15% risk delta).
- **Customs / Public Holidays:** Evaluates destination country holiday calendars within a ±2 day ETA window (+5% risk delta).
- **Foreign Exchange Volatility:** Monitors currency shifts (e.g. USD/INR) that trigger customs clearance delays (+2% risk delta).

C. Generative AI Logistics Co-Pilot

Integrates Google Gemini & NVIDIA NIM to translate complex statistical variables into clear operational insights. Operators can perform natural language Q&A;, generate delay notification emails to consignees with one click, and obtain instant 2-step actionable mitigation plans. Includes an autonomous fallback rule engine for 100% uptime.

5. Mathematical Formulation & Scoring Algorithm

1. Point-in-Time Historical Evaluation (Zero Data Leakage):
To ensure models only learn from historical reality, all calculations query exclusively records completed before the shipment's ETD:

```
Route_Delay_As_Of(T) = AVG(delay_days) WHERE event_timestamp < ETD
Carrier_Reliability_As_Of(T) = (On_Time_Shipments < ETD / Total_Shipments < ETD) * 100
```

2. Multi-Factor Risk Score Equation:
The raw input vector *z* combines normalized carrier reliability, historical lane delay, seasonal surges, cargo class, and transit variance:

```
z = (70 - Carrier_Reliability)/18 + (Route_Delay)/3.5 + W_season + W_cargo + W_mode + max(0,
-Transit_Delta)/4 - 1.2
```

The base score is mapped to a probability via the Sigmoid function and adjusted with external real-world signals:

```
Base_Score = 1 / (1 + exp(-z))
Final_Risk_Score = min(0.99, max(0.01, Base_Score + Sum(Delta_External)))
```

6. Database Architecture & Schema Design

The relational database is normalized in 3NF across 6 core domain entities:

Entity Table	Primary Key	Key Attributes & Relationships
CARRIERS	`carrier_id`	carrier_code, carrier_name, on_time_pct_hist. Tracks long-term carrier reliability.
ROUTES	`route_id`	origin_port, dest_port, mode, avg_transit_days. Trade lane baseline transit profiles.
SHIPMENTS	`shipment_id`	shipment_ref, carrier_id, route_id, etd, eta, actual_arrival, cargo_type, status. Active freight worklist.
RISK_SCORES	`shipment_id (1:1)`	risk_score, risk_tier, top_factors (JSON), recommendation, scored_at. Live scoring cache.
SHIPMENT_HISTORY	`history_id`	shipment_id, event_type, delay_days, event_ts. Historical audit trail for time-series analytics.
EXTERNAL_TELEMETRY	`id`	Port weather conditions, terminal wait hours, holiday calendars, and FX exchange rates.